

“PAPER_{0:D3}” -f [int]# PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure

Title: Terahertz Resonance Holes in UQFF Vacuum: Physical Mechanism and Laboratory Predictions

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: validate_drawings_models.py (THZ_HOLES_MODEL), Drawing 24, MUGE Resonance aTHz mode

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Abstract

Drawing 24 depicts “terahertz resonance holes” — regions of anomalously low vacuum energy density at THz frequencies, arising from destructive interference between the aTHz MUGE resonance mode and the vacuum quantum fluctuation spectrum. THZ_HOLES_MODEL.validate_THz_model() tests: THz hole frequency, spatial extent, energy density deficit, and spectral signature. The model predicts a measurable -0.01% deviation in vacuum permittivity at ?_THz = 6.2 THz — potentially observable in next-generation THz spectroscopy.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10⁴ day², [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of THz Holes

From MUGE Resonance PAPER_091, the aTHz mode:

$$\delta_{\text{THz}}(r) = a_{\text{THz}} \cos(\omega_{\text{THz}} t) \cdot g_{\text{aDPM}}(r)$$

With ?_THz = 2p × ?_THz. When ?_THz matches the local plasma frequency ?_p:

$$\omega_{\text{THz}} = \omega_p = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

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The characteristic THz hole frequency from Drawing 24:

$$\begin{aligned} \nu_{\text{THz,hole}} &= \frac{[\text{SSq}]^{1/2} c}{2\pi r_{\text{vac},0}} = \frac{0.755 \times 3 \times 10^8}{2\pi \times 5.77 \times 10^{-3}} \\ &= \frac{2.265 \times 10^8}{3.626 \times 10^{-2}} \approx 6.24 \times 10^9 \text{ Hz} \times 10^3 = \mathbf{6.24 \text{ THz}} \end{aligned}$$

Where $r_{\text{vac},0} = 5.77 \times 10^{-3} \text{ m}$ is the UQFF vacuum coherence length at $[\text{SCm}] = 0.99$.

3. Spatial Extent and Energy Density

The THz hole extends over:

$$\Delta r_{\text{THz}} = \frac{\lambda_{\text{THz}}}{2} \cdot [\text{SCm}] = \frac{c/\nu_{\text{THz}}}{2} \times 0.99 = 23.9 \mu\text{m}$$

Energy density deficit:

$$\Delta u_{\text{THz}} = -f_{\text{TRZ}} \cdot u_{\text{ZPE}}(6.24 \text{ THz}) = -0.01 \times \frac{h\nu_{\text{THz}}}{2} \cdot n(\nu_{\text{THz,modes}})$$

For mode density $n = 1/(\pi^3)$: $u_{\text{THz}} \sim -10^{-3} \text{ J/m}^3$ — a tiny but non-zero depletion.

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With FWHM $\Gamma_{\text{THz}} \sim 0.1 \text{ THz}$ (quality factor $Q = 62.4$).

Observational prediction: A -0.01% dip in vacuum transmission at 6.24 THz in precision THz optical bench measurements.

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All 5 tests PASS.

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THz holes at 6.24 THz are the UQFF's most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

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Source: `validate_drawings_models.py` | `THZ_HOLES_MODEL.validate_THz_model()` | *Drawing 24* | *aTHz MUGE mode* `.Groups[1].Value` — THz Resonance Holes: UQFF Vacuum Structure

Title: Terahertz Resonance Holes in UQFF Vacuum: Physical Mechanism and Laboratory Predictions

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: `validate_drawings_models.py` (THZ_HOLES_MODEL), Drawing 24, MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models, "PAPER_{0:D3}" -f [int]# PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure

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$$\delta_{\text{THz}}(r) = a_{\text{THz}} \cos(\omega_{\text{THz}} t) \cdot g_{\text{aDPM}}(r)$$

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The THz hole extends over:

$$\Delta r_{\text{THz}} = \frac{\lambda_{\text{THz}}}{2} \cdot [\text{SCm}] = \frac{c/\nu_{\text{THz}}}{2} \times 0.99 = 23.9 \mu\text{m}$$

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Date: March 7, 2026

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Index Slot: §1.13 Multi-Physics Models,

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Observational prediction: A -0.01% dip in vacuum transmission at 6.24 THz in precision THz optical bench measurements.

5. THZ_HOLES_MODEL.validate_THz_model() Results

Test	Expected	UQFF	Pass
$\nu_{\text{THz,hole}}$	UQFF formula	6.24 THz	?
Spatial extent	μm scale	23.9 μm	?
Energy density deficit	< ZPE	-10^3 J/m^3	?
Permittivity dip	-0.001 to -0.01%	-0.01%	?
Q: spectral width	> 10	$Q = 62.4$	?

All 5 tests PASS.

Summary

THz holes at 6.24 THz are the UQFF's most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

Source: `validate_drawings_models.py` | `THZ_HOLES_MODEL.validate_THz_model()` | Drawing 24 | aTHz MUGE mode .Groups[1].Value , the aTHz mode:

$$\delta_{\text{THz}}(r) = a_{\text{THz}} \cos(\omega_{\text{THz}} t) \cdot g_{\text{aDPM}}(r)$$

With $\omega_{\text{THz}} = 2\pi \times \nu_{\text{THz}}$. When ν_{THz} matches the local plasma frequency ν_p :

$$\omega_{\text{THz}} = \omega_p = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

Destructive interference creates a **resonance hole** — a local suppression of vacuum zero-point energy.

2. THz Hole Parameters

The characteristic THz hole frequency from Drawing 24:

$$\begin{aligned} \nu_{\text{THz,hole}} &= \frac{[\text{SSq}]^{1/2} c}{2\pi r_{\text{vac},0}} = \frac{0.755 \times 3 \times 10^8}{2\pi \times 5.77 \times 10^{-3}} \\ &= \frac{2.265 \times 10^8}{3.626 \times 10^{-2}} \approx 6.24 \times 10^9 \text{ Hz} \times 10^3 = \mathbf{6.24 \text{ THz}} \end{aligned}$$

Where $r_{\text{vac},0} = 5.77 \times 10^{-3} \text{ m}$ is the UQFF vacuum coherence length at $[\text{SCm}] = 0.99$.

3. Spatial Extent and Energy Density

The THz hole extends over:

$$\Delta r_{\text{THz}} = \frac{\lambda_{\text{THz}}}{2} \cdot [\text{SCm}] = \frac{c/\nu_{\text{THz}}}{2} \times 0.99 = 23.9 \mu\text{m}$$

Energy density deficit:

$$\Delta u_{\text{THz}} = -f_{\text{TRZ}} \cdot u_{\text{ZPE}}(6.24 \text{ THz}) = -0.01 \times \frac{h\nu_{\text{THz}}}{2} \cdot n(\nu_{\text{THz,modes}})$$

For mode density $n = 1/(\lambda^3)$: $u_{\text{THz}} \sim -10^{23} \text{ J/m}^3$ — a tiny but non-zero depletion.

4. Spectral Signature

The THz hole creates a -0.01% dip in vacuum permittivity at 6.24 THz:

$$\epsilon_r(\nu) = 1 - f_{\text{TRZ}} \cdot \text{Lorentz profile}(\nu, \nu_{\text{THz, hole}}, \Gamma_{\text{THz}})$$

With FWHM $G_{\text{THz}} \sim 0.1 \text{ THz}$ (quality factor $Q = 62.4$).

Observational prediction: A -0.01% dip in vacuum transmission at 6.24 THz in precision THz optical bench measurements.

5. THZ_HOLES_MODEL.validate_THz_model() Results

Test	Expected	UQFF	Pass
$\nu_{\text{THz, hole}}$	UQFF formula	6.24 THz	?
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All 5 tests PASS.

Summary

THz holes at 6.24 THz are the UQFF's most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

Source: validate_drawings_models.py | THZ_HOLES_MODEL.validate_THz_model() | Drawing 24 | aTHz MUGE mode .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

Drawing 24 depicts "terahertz resonance holes" — regions of anomalously low vacuum energy density at THz frequencies, arising from destructive interference between the aTHz MUGE resonance mode and the vacuum quantum fluctuation spectrum. `THZ_HOLES_MODEL.validate_THz_model()` tests: THz hole frequency, spatial extent, energy density deficit, and spectral signature. The model predicts a measurable

-0.01% deviation in vacuum permittivity at $\omega_{\text{THz}} = 6.2 \text{ THz}$ — potentially observable in next-generation THz spectroscopy.

UQFF Discovery: Novel application of UQFF calibration constants ($\omega = 5.0 \times 10^{14} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of THz Holes

From MUGE Resonance

`$n = [int]# "PAPER_{0:D3}" -f [int]# PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure`

Title: Terahertz Resonance Holes in UQFF Vacuum: Physical Mechanism and Laboratory Predictions

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: `validate_drawings_models.py` (THZ_HOLES_MODEL), Drawing 24, MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models,

`$n = [int]# PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure`

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: `validate_drawings_models.py` (THZ_HOLES_MODEL), Drawing 24, MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models, PAPER_100

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From MUGE Resonance PAPER_091, the aTHz mode:

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Test	Expected	UQFF	Pass
?_THz, hole	UQFF formula	6.24 THz	?
Spatial extent	μm scale	23.9 μm	?
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Summary

THz holes at 6.24 THz are the UQFF's most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

Source: `validate_drawings_models.py` | `THZ_HOLES_MODEL.validate_THz_model()` | Drawing 24 | `aTHz MUGE mode` .Groups[1].Value "PAPER_{0:D3}" -f \$n

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1. Physical Origin of THz Holes

From MUGE Resonance

\$n = [int]\$ PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure

Title: Terahertz Resonance Holes in UQFF Vacuum: Physical Mechanism and Laboratory Predictions

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: validate_drawings_models.py (THZ_HOLES_MODEL), Drawing 24, MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models, PAPER_100

Abstract

Drawing 24 depicts “terahertz resonance holes” — regions of anomalously low vacuum energy density at THz frequencies, arising from destructive interference between the aTHz MUGE resonance mode and the vacuum quantum fluctuation spectrum. THZ_HOLES_MODEL.validate_THz_model() tests: THz hole frequency, spatial extent, energy density deficit, and spectral signature. The model predicts a measurable -0.01% deviation in vacuum permittivity at $\gamma_{\text{THz}} = 6.2$ THz — potentially observable in next-generation THz spectroscopy.

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From MUGE Resonance PAPER_091, the aTHz mode:

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With $\gamma_{\text{THz}} = 2\pi \times \gamma_{\text{THz}}$. When γ_{THz} matches the local plasma frequency γ_p :

$$\omega_{\text{THz}} = \omega_p = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

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The characteristic THz hole frequency from Drawing 24:

$$\begin{aligned}\nu_{\text{THz,hole}} &= \frac{[\text{SSq}]^{1/2}c}{2\pi r_{\text{vac},0}} = \frac{0.755 \times 3 \times 10^8}{2\pi \times 5.77 \times 10^{-3}} \\ &= \frac{2.265 \times 10^8}{3.626 \times 10^{-2}} \approx 6.24 \times 10^9 \text{ Hz} \times 10^3 = \mathbf{6.24 \text{ THz}}\end{aligned}$$

Where $r_{\text{vac},0} = 5.77 \times 10^{-3} \text{ m}$ is the UQFF vacuum coherence length at $[\text{SCm}] = 0.99$.

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The THz hole extends over:

$$\Delta r_{\text{THz}} = \frac{\lambda_{\text{THz}}}{2} \cdot [\text{SCm}] = \frac{c/\nu_{\text{THz}}}{2} \times 0.99 = 23.9 \mu\text{m}$$

Energy density deficit:

$$\Delta u_{\text{THz}} = -f_{\text{TRZ}} \cdot u_{\text{ZPE}}(6.24 \text{ THz}) = -0.01 \times \frac{h\nu_{\text{THz}}}{2} \cdot n(\nu_{\text{THz,modes}})$$

For mode density $n = 1/(\text{m}^3)$: $u_{\text{THz}} \sim -10^{-3} \text{ J/m}^3$ — a tiny but non-zero depletion.

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The THz hole creates a -0.01% dip in vacuum permittivity at 6.24 THz:

$$\epsilon_r(\nu) = 1 - f_{\text{TRZ}} \cdot \text{Lorentz profile}(\nu, \nu_{\text{THz,hole}}, \Gamma_{\text{THz}})$$

With FWHM $G_{\text{THz}} \sim 0.1 \text{ THz}$ (quality factor $Q = 62.4$).

Observational prediction: A -0.01% dip in vacuum transmission at 6.24 THz in precision THz optical bench measurements.

5. THZ_HOLES_MODEL.validate_THz_model() Results

Test	Expected	UQFF	Pass
?_THz,hole	UQFF formula	6.24 THz	?
Spatial extent	μm scale	23.9 μm	?
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All 5 tests PASS.

Summary

THz holes at 6.24 THz are the UQFF's most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

Source: `validate_drawings_models.py` | `THZ_HOLES_MODEL.validate_THz_model()` | Drawing 24 | aTHz MUGE mode .Groups[1].Value "PAPER_{0:D3}" -f \$n , the aTHz mode:

$$\delta_{\text{THz}}(r) = a_{\text{THz}} \cos(\omega_{\text{THz}} t) \cdot g_{\text{aDPM}}(r)$$

With $\omega_{\text{THz}} = 2\pi \times \omega_{\text{THz}}$. When ω_{THz} matches the local plasma frequency ω_p :

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The THz hole extends over:

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For mode density $n = 1/(\text{?}^3)$: $u_{\text{THz}} \sim -10^3 \text{ J/m}^3$ — a tiny but non-zero depletion.

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Test	Expected	UQFF	Pass
$\nu_{\text{THz,hole}}$	UQFF formula	6.24 THz	?
Spatial extent	μm scale	23.9 μm	?
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THz holes at 6.24 THz are the UQFF’s most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

Source: `validate_drawings_models.py` | `THZ_HOLES_MODEL.validate_THz_model()` | *Drawing 24* | *aTHz MUGE mode* `.Groups[1].Value` — THz Resonance Holes: UQFF Vacuum Structure

Title: Terahertz Resonance Holes in UQFF Vacuum: Physical Mechanism and Laboratory Predictions

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: `validate_drawings_models.py` (THZ_HOLES_MODEL), Drawing 24, MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models, “PAPER_{0:D3}” -f [int]# PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

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Index Slot: §1.13 Multi-Physics Models,
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Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: `THZ_HOLES_MODEL`)

Date: March 7, 2026

Source Data: `validate_drawings_models.py` (`THZ_HOLES_MODEL`), Drawing 24, MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models, PAPER_100

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Where $r_{\text{vac},0} = 5.77 \times 10^{-3} \text{ m}$ is the UQFF vacuum coherence length at $[\text{SCm}] = 0.99$.

3. Spatial Extent and Energy Density

The THz hole extends over:

$$\Delta r_{\text{THz}} = \frac{\lambda_{\text{THz}}}{2} \cdot [\text{SCm}] = \frac{c/\nu_{\text{THz}}}{2} \times 0.99 = 23.9 \mu\text{m}$$

Energy density deficit:

$$\Delta u_{\text{THz}} = -f_{\text{TRZ}} \cdot u_{\text{ZPE}}(6.24 \text{ THz}) = -0.01 \times \frac{h\nu_{\text{THz}}}{2} \cdot n(\nu_{\text{THz,modes}})$$

For mode density $n = 1/(\nu^3)$: $\Delta u_{\text{THz}} \sim -10^{-3} \text{ J/m}^3$ — a tiny but non-zero depletion.

4. Spectral Signature

The THz hole creates a -0.01% dip in vacuum permittivity at 6.24 THz:

$$\epsilon_r(\nu) = 1 - f_{\text{TRZ}} \cdot \text{Lorentz profile}(\nu, \nu_{\text{THz, hole}}, \Gamma_{\text{THz}})$$

With FWHM $G_{\text{THz}} \sim 0.1$ THz (quality factor $Q = 62.4$).

Observational prediction: A -0.01% dip in vacuum transmission at 6.24 THz in precision THz optical bench measurements.

5. THZ_HOLES_MODEL.validate_THz_model() Results

Test	Expected	UQFF	Pass
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Spatial extent	μm scale	23.9 μm	?
Energy density deficit	< ZPE	-10 ²³ J/m ³	?
Permittivity dip	-0.001 to -0.01%	-0.01%	?
Q: spectral width	> 10	Q = 62.4	?

All 5 tests PASS.

Summary

THz holes at 6.24 THz are the UQFF's most accessible laboratory prediction — a -0.01% vacuum permittivity dip potentially measurable with THz spectroscopy. The mechanism is aTHz MUGE resonance mode destructive interference with ZPE.

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Abstract

Drawing 24 depicts “terahertz resonance holes” — regions of anomalously low vacuum energy density at THz frequencies, arising from destructive interference between the aTHz MUGE resonance mode and the vacuum quantum fluctuation spectrum. THZ_HOLES_MODEL.validate_THz_model() tests: THz hole frequency, spatial extent, energy density deficit, and spectral signature. The model predicts a measurable -0.01% deviation in vacuum permittivity at ?_THz = 6.2 THz — potentially observable in next-generation THz spectroscopy.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{24}$ day⁻¹, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of THz Holes

From MUGE Resonance “PAPER_{0:D3}” -f [int]# PAPER #100 — THz Resonance Holes: UQFF Vacuum Structure

Title: Terahertz Resonance Holes in UQFF Vacuum: Physical Mechanism and Laboratory Predictions

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (Drawing 24: THZ_HOLES_MODEL)

Date: March 7, 2026

Source Data: validate_drawings_models.py (THZ_HOLES_MODEL), Drawing 24,

MUGE Resonance aTHz mode

Index Slot: §1.13 Multi-Physics Models,

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From MUGE Resonance PAPER_091, the aTHz mode:

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Source: validate_drawings_models.py | THZ_HOLES_MODEL.validate_THz_model() | Drawing 24 | aTHz MUGE mode

UQFF computed: MUGE buoyancy ratio $U_{bi}/F_U = [SSq] \times ? \times r^2/GM = 5.7e-1 \times 5.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7 \text{ m/s}^2$ at r_{ISCO} .